

ANATOMY OF FLOWERING PLANTS

Hierarchy of Organization: Plants have cells as the basic unit, which are organized into tissues, and tissues are organized into organs.

- *Organ and Class Differences:* Different organs in a plant show differences in their internal structure. Within angiosperms, monocots and dicots are also seen to be anatomically different.
- *Adaptation:* Internal structures also show adaptations to diverse environments.
- *Tissue Functions:* The main functions of plant tissues include assimilation and storage of food, transportation of water, minerals and photosynthates, and mechanical support.

II. The Tissue System

Based on their structure and location, there are three types of tissue systems in the plant body.

I. Epidermal Tissue System (ETS)

The epidermal tissue system forms the outermost covering of the entire plant body. It comprises epidermal cells, stomata, and epidermal appendages (trichomes and hairs).

- *Epidermis:* This is the outermost layer of the primary plant body. It is usually single-layered and made up of elongated, compactly arranged cells. Epidermal cells are parenchymatous, containing a large vacuole and a small amount of cytoplasm lining the cell wall.

Cuticle: The outside of the epidermis is often covered with a waxy thick layer called the cuticle, which prevents the loss of water. The cuticle is absent in roots.

- **Root Hairs:** These are unicellular elongations of the epidermal cells and function to help absorb water and minerals from the soil.

- **Trichomes (Epidermal Hairs on Stem):** These are epidermal hairs in the shoot system, usually multicellular. They may be branched or unbranched, soft or stiff, and may even be secretory. Trichomes help in preventing water loss due to transpiration.

- **Stomata:** These structures are present in the epidermis of leaves and regulate the processes of transpiration and gaseous exchange.

- **Guard Cells:** Each stoma is composed of two bean-shaped cells known as guard cells which enclose the stomatal pore. In grasses, the guard cells are dumb-bell shaped.

- **Wall Structure:** The outer walls of guard cells (away from the pore) are thin, while the inner walls (towards the pore) are highly thickened. Guard cells possess chloroplasts and regulate the opening and closing of stomata.

- **Subsidiary Cells:** Sometimes, a few epidermal cells near the guard cells become specialized in shape and size and are called subsidiary cells.

- **Stomatal Apparatus:** This term refers collectively to the stomatal aperture (pore), the guard cells, and the

- Arrangement Types:

- Radial Arrangement: Xylem and phloem are arranged in an alternate manner along different radii, typically found in roots.

- Conjoint Arrangement: Xylem and phloem are jointly situated along the same radius of the vascular bundle. This is common in stems and leaves. In conjoint bundles, the phloem is usually located only on the outer side of the xylem.

III. Anatomy of Dicotyledonous and Monocotyledonous Organs

The tissue organization is best understood by studying the transverse sections of the mature zones of roots, stems, and leaves.

1. Dicotyledonous Root (e.g., Sunflower root)

- Outermost Layer: The outermost layer is epiblema. Many cells of the epiblema protrude as unicellular root hairs.

- Cortex: Consists of several layers of thin-walled parenchyma cells with intercellular spaces.

- Endodermis: The innermost layer of the cortex. It comprises a single layer of barrel-shaped cells without intercellular spaces. The tangential and radial walls of the endodermal cells have a deposition of water-impermeable, waxy material called suberin in the form of casparian strips.

- Pericycle: A few layers of thick-walled parenchymatous cells lying next to the endodermis. The initiation of lateral roots and vascular cambium during secondary growth takes

place in these cells.

- **Vascular Bundles:** There are usually two to four patches of xylem and phloem.
- **Conjunctive Tissue:** The parenchymatous cells situated between the xylem and the phloem.
- **Pith:** Small or inconspicuous.
- **Stele:** All tissues on the inner side of the endodermis, including pericycle, vascular bundles, and pith, constitute the stele.
- **Secondary Growth:** A cambium ring develops later between the xylem and phloem. Secondary growth occurs in most dicotyledonous roots.

2. Monocotyledonous Root:-

The anatomy of the monocot root is similar to the dicot root in several respects, having epidermis, cortex, endodermis, pericycle, vascular bundles, and pith.

- **Vascular Bundles:** Monocot roots typically have more than six (polyarch) xylem bundles, unlike the fewer bundles in dicot roots.
- **Pith:** The pith is large and well developed.
- **Secondary Growth:** Monocotyledonous roots do not undergo any secondary growth.

3. Dicotyledonous Stem (Young, Typical):-

- **Epidermis:** The outermost protective layer. Covered with a

- **Cortex:** Arranged in multiple layers between the epidermis and pericycle, consisting of three sub-zones.
 - **Hypodermis:** The outer sub-zone, consisting of a few layers of collenchymatous cells located just below the epidermis, providing mechanical strength to the young stem.
 - **Cortical Layers:** Below the hypodermis, consisting of rounded, thin-walled parenchymatous cells with conspicuous intercellular spaces.
 - **Endodermis:** The innermost layer of the cortex, also referred to as the starch sheath because its cells are rich in starch grains.
- **Pericycle:** Present on the inner side of the endodermis and above the phloem, structured as semi-lunar patches of sclerenchyma.
- **Medullary Rays:** A few layers of radially placed parenchymatous cells located between the vascular bundles.
- **Vascular Bundles:** A large number of vascular bundles are arranged in a ring, which is a characteristic feature of the dicot stem. Each bundle is conjoint, open, and possesses endarch protoxylem.
- **Pith:** A large number of rounded, parenchymatous cells with large intercellular spaces occupy the central portion of the stem, constituting the pith.

4. Monocotyledonous Stem:-

- **Hypodermis:** Sclerenchymatous.

- **Bundle Sheath:** Each vascular bundle is surrounded by a sclerenchymatous bundle sheath.
- **Arrangement and Structure:** Vascular bundles are conjoint and closed. Peripheral bundles are generally smaller than the centrally located ones.
- **Phloem Parenchyma:** Absent.
- **Cavities:** Water-containing cavities are present within the vascular bundles.

5. Dorsiventral (Dicotyledonous) Leaf:-

The leaf lamina's vertical section shows three main parts: epidermis, mesophyll, and vascular system.

- **Epidermis:** Covers both the upper (adaxial) and lower (abaxial) surfaces and has a conspicuous cuticle. The abaxial epidermis generally bears more stomata than the adaxial epidermis, which may even lack stomata.
- **Mesophyll:** The tissue between the upper and lower epidermis. It possesses chloroplasts, carries out photosynthesis, and is made up of parenchyma. It is differentiated into two types of cells:
 - **Palisade Parenchyma:** Adaxially placed, made up of elongated cells arranged vertically and parallel to each other.
 - **Spongy Parenchyma:** Located below the palisade cells, consisting of oval or round and loosely arranged cells that extend to the lower epidermis. There are numerous large spaces and air cavities between these cells.

- **Vascular System:** Vascular bundles are visible in the veins and midrib. The size of the bundles depends on the size of the veins, which vary in thickness due to the reticulate venation characteristic of dicot leaves. The vascular bundles are surrounded by a layer of thick walled bundle sheath cells.

6. Isobilateral (Monocotyledonous) Leaf:-

The anatomy is similar to the dorsiventral leaf in many ways, but displays characteristic differences:

- **Stomata:** Stomata are present on both the adaxial and abaxial surfaces of the epidermis.
- **Mesophyll Differentiation:** The mesophyll is not differentiated into palisade and spongy parenchyma.
- **Bulliform Cells (in Grasses):** Certain adaxial epidermal cells along the veins modify into large, empty, colorless cells. When these cells absorb water and become turgid, the leaf surface is exposed. When they are flaccid due to water stress, they cause the leaves to curl inwards to minimize water loss.
- **Venation Reflection:** The parallel venation typical of monocot leaves is reflected in the near similar sizes of vascular bundles (except in main veins).

IV. Summary of Differences

Monocotyledonous and dicotyledonous plants show marked variation in their internal structures. They differ in the type, number, and location of vascular bundles. Secondary growth occurs in most dicotyledonous roots and stems, but not in

2. Ground Tissue System (GTS)

All tissues except the epidermis and vascular bundles constitute the ground tissue. This system forms the main bulk of the plant.

- **Composition:** It consists of simple tissues such as parenchyma, collenchyma, and sclerenchyma.
- **Location:** Parenchymatous cells are typically present in the cortex, pericycle, pith, and medullary rays in the primary stems and roots.
- **In Leaves:** The ground tissue in leaves consists of thin-walled, chloroplast-containing cells and is termed the mesophyll.
- **Zones:** The ground tissue system is generally divided into three major zones: cortex, pericycle, and pith.

3. Vascular Tissue System (VTS)

The vascular system consists of complex tissues, the phloem and the xylem. Xylem and phloem together constitute vascular bundles. The vascular bundles form the conducting tissue, translocating water, minerals, and food material.

• Open vs. Closed Bundles:

- **Open Vascular Bundles:** Found in dicotyledonous stems, cambium is present between phloem and xylem. Due to cambium, these bundles possess the ability to form secondary xylem and phloem tissues.
- **Closed Vascular Bundles:** Found in monocotyledons,

monocot roots.

For instance, a plant material can be identified as a monocot stem if it shows conjoint, scattered vascular bundles surrounded by a sclerenchymatous bundle sheath, and if phloem parenchyma is absent.

